Mapping of X/Z Basis Protocol to Angles

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This document outlines the mapping between a time-bin phase encoding quantum key distribution (QKD) protocol and the numerical angles used with VeriQloud hardware.

In the X/Z basis protocol, Alice prepares a quantum state in one of two bases (Z or X), and Bob performs a measurement by randomly choosing one of the same two bases. This document explicits the mapping between Alice's state preparation and Bob's measurement choice into numerical angles.

Mapping the X/Z Protocol to Angles

The physical protocol is based on time-bin phase encoding. Alice can prepare four different states, corresponding to two states in each of the two bases, Z and X. Bob measures by choosing a basis, Z or X. These actions can be mapped to numerical angles.

Alice's State Preparation

Alice prepares one of four states. These states and their corresponding angle representations are:

- **Z Basis**:
  - *Early* time-bin state: mapped to angle **0**.
  - *Late* time-bin state: mapped to angle **64**.
- **X Basis**:
  - *0-phase shift* state: mapped to angle **32**.
  - Π-*phase shift* state: mapped to angle **96**.

Alice's choice of basis is random, and for a given basis, her choice of state can be used to encode a key bit. Her state preparation is represented by an angle $A \in \{0, 32, 64, 96\}$.

Bob's Measurement

Bob randomly chooses to measure in either the Z or X basis. In the physical protocol, Bob does not apply an additional operation corresponding to a bit flip (Pauli-X gate). Bob only have the two basis choices with their corresponding angles :

- Bob choosing the **Z basis** is represented by angle **0**.
- Bob choosing the **X basis** is represented by angle **32**.

The mapping tables in this document show the complete set of outcomes for all four possible angles for Bob for completeness. However, only the rows where Bob's angle is 0 or 32 are physically reachable in this protocol.

Mapping Angles to Protocol Outcome

The outcome of each event is determined by a probability derived from the sum of Alice's angle (A) and Bob's angle (B). A total angle θ is calculated, which is then used to look up a probability value. The outcomes in the tables are consistent with a probability for outcome '1' given by the formula:

$$P(1) = \sin^2\left(\frac{\theta \cdot \pi}{128}\right)$$

The total angle θ calculation can include an offset, which emulates different initial quantum states. The general formula for the total angle is $\theta = (A + B + \text{offset}) \pmod{128}$.

When the measurement bases of Alice and Bob align, the result is deterministic (correlated). This occurs when $P(1)$ is either 0 or 1.

- **Correlated (Outcome 0)**: $\theta = 0$, which means $P(1) = \sin^2(0) = 0$.
- **Correlated (Outcome 1)**: $\theta = 64$, which means $P(1) = \sin^2(\pi/2) = 1$.

When their bases are misaligned, the result is random.

- **Random**: $\theta = 32$ or 96, which means $P(1) = \sin^2(\pi/4) = 1/2$.

Mapping Tables

The following tables detail the outcomes for different angle choices by Alice and Bob, both with and without an initial state offset.

Without Offset (Simulating Initial State $|0\rangle$)

This table shows the hypothetical outcomes if the formula were $\theta = (A + B) \pmod{128}$. This simulates a scenario where the initial quantum state is $|0\rangle$.

Table 1: Protocol outcomes without an offset.

Alice's Angle	Bob's Angle	Alice's Basis	Bob's Basis	(A+B)&127	Result Type	Determ. Outcome
0	0	Z	Z	0	Correlated	0
0	32	Z	X	32	Random	-
0	64	Z	Z	64	Correlated	1
0	96	Z	X	96	Random	-
32	0	X	Z	32	Random	-
32	32	X	X	64	Correlated	1
32	64	X	Z	96	Random	-
32	96	X	X	0	Correlated	0
64	0	Z	Z	64	Correlated	1
64	32	Z	X	96	Random	-
64	64	Z	Z	0	Correlated	0
64	96	Z	X	32	Random	-
96	0	X	Z	96	Random	-
96	32	X	X	0	Correlated	0
96	64	X	Z	32	Random	-
96	96	X	X	64	Correlated	1

With +32 Offset (Simulating Initial State $|+\rangle$)

This table shows the outcomes based on the formula $\theta = (A+B+32) \pmod{128}$. This offset is used to simulate a scenario where the initial quantum state is $|+\rangle$.

Table 2: Protocol outcomes with a +32 offset.

Alice's Angle	Bob's Angle	Alice's Basis	Bob's Basis	(A+B+32)&127	Result Type	Determ. Outc
0	0	Z	Z	32	Random	-
0	32	Z	X	64	Correlated	1
0	64	Z	Z	96	Random	-
0	96	Z	X	0	Correlated	0
32	0	X	Z	64	Correlated	1
32	32	X	X	96	Random	-
32	64	X	Z	0	Correlated	0
32	96	X	X	32	Random	-
64	0	Z	Z	96	Random	-
64	32	Z	X	0	Correlated	0
64	64	Z	Z	32	Random	-
64	96	Z	X	64	Correlated	1
96	0	X	Z	0	Correlated	0
96	32	X	X	32	Random	-
96	64	X	Z	64	Correlated	1
96	96	X	X	96	Random	-

Data Interface and Assumptions

The post-processing software makes several key assumptions about the data it receives from the FIFOs:

- **Event Validity:** The software assumes that all events received through the FIFOs are valid, single-photon detection events. Issues like photon loss (no click) or dark counts would desynchronize the streams. There are expected to be filtered out by upstream hardware or a pre-processing layer. If you do not want to discard these events (for security reasons as you mentionned), you shall add random data at the same index so the streams must still remain synchronized.
- **Synchronization:** It is assumed that the data streams for Alice and Bob are perfectly synchronized. Each event (a timestamped angle from Alice and a timestamped GCR from Bob) corresponds to the same quantum event. The post-processing relies on this one-to-one correspondence.
- **Angle Indexing:** The angle FIFOs do not contain raw angle values but rather their corresponding indices (0-3), written as two bits. The mapping is as follows:
  - Angle 0 is mapped to index 0 (two bits value 00).
  - Angle 32 is mapped to index 1 (two bits value 01).
  - Angle 64 is mapped to index 2 (two bits value 10).
  - Angle 96 is mapped to index 3 (two bits value 11).

  The angles are then packed two by two as stated in previous document.
- **Clarification on Bob's Data:** Bob's measurement result is not encoded as an angle.
  - Bob's **basis choice** is encoded as an angle (0 for Z, 32 for X). This is written to his angle FIFO.
  - Bob's **measurement outcome** (the logical 0 or 1 result from his detectors) is the click-result 'R' part of the GCR data written to the GCR FIFO.

  For example, if Alice prepares the *Late* Z-basis state (angle 64) and Bob measures in the Z-basis (angle 0), they have chosen the same basis. The outcome is a deterministic '1', which Bob records in his GCR stream.

The fifo path should be specified by a configuration file to allow running multiple instances of the programs on the same machine.